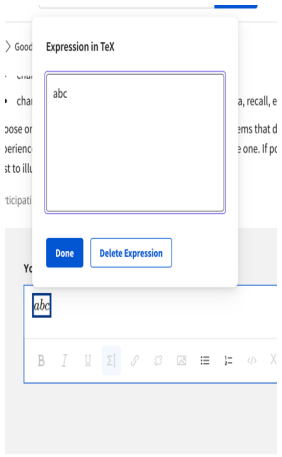
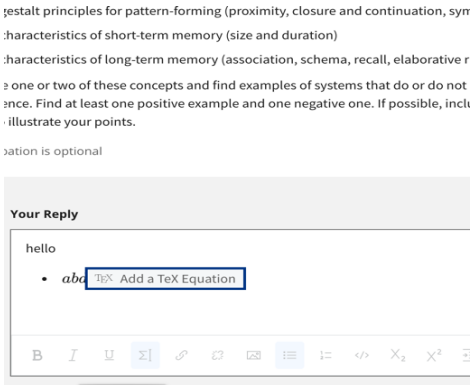
Items	Description	System Location	Heuristics Violated	Severity
Item 1	There is no **undo/redo** button. Most users expect there to be an undo/redo option when typing paragraphs of text with text formatting options such as bold, italicize, underline, bullet, etc. This should be an option as it would be consistent with standards provided on most discussion board interfaces as well as Microsoft word, email, and other text-focused interfaces.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6	#4 Consistency and Standards #3 User Control and Freedom	3
Item 2	If you type text in the discussion prompt box and refresh the page or leave the discussion prompt page to go to a different part of the course, any **text typed in the discussion box will not be saved** from the previous visit. There should be a popup to prevent errors asking if the user is sure they want to leave and warning them that their work will not be saved.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6	#5 Error Prevention	3
Item 3	When **replying to a discussion board post**, if the post has multiple replies already, the **text box for submitting a reply appears at the end** of the long thread of replies. A typical user might only want to read through the original post and share their thoughts on that. However, they have to scroll past every reply to find the box where they can submit their own reply. Instead, the replies should be automatically hidden with an option saying "show *x* replies." There should be a text box available immediately following the original poster's words so the user can easily reference the original poster's thoughts as they form their own reply. This will also make scrolling through responses more efficient.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 --> replying to a discussion board post	#2 Match Between System and Real World #7 Flexibility and Efficiency of Use	2
Item 4	When interacting with discussion board responses, the term "***x* Upvotes**" is not consistent with terminology used on most platforms. Most users are familiar with the term "like." It is commonly coupled with an icon such as a heart or a thumbs-up. The word "like" is an actionable step the user could take. The term "*x* Upvotes" is a more passive phrase that connotates a statistic rather than an actionable step. The thumbs-up "Like" button would be more consistent as it appears at the bottom of all other items on the course website besides the Discussion Prompt page.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 --> interacting with a discussion board post	#2 Match Between System and Real World #4 Consistency and Standards	2

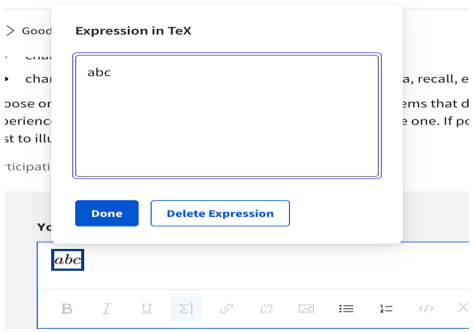
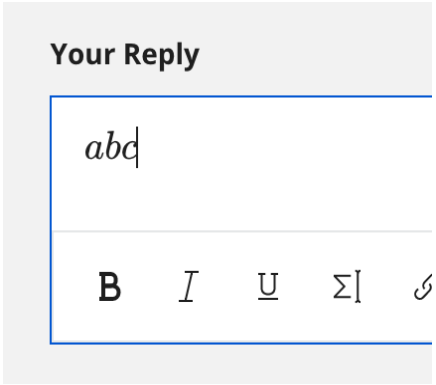
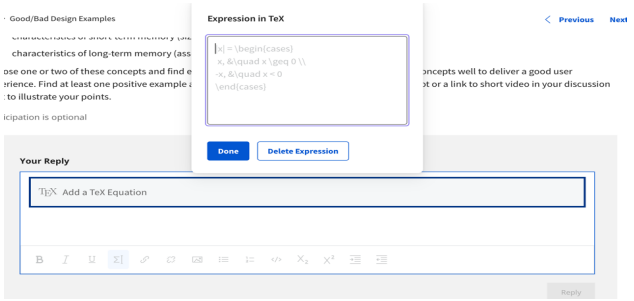
Item 5	The word "**reply**" is confusing to a user when it come to submitting an orginal post to the discussion board. To many users "**submit response**" would be more familiar and consistent with interactions users have had with other discussion-board interfaces. Additionally, the user receives a message that states "Your response has been submitted. Engage and discuss with other learners below!" After clicking "reply." The word "reply" should be saved for when a user is submitting a reply to another user's initial response.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6	#4 Consistency and Standards	2
Item 6	When **resizing an inserted image**, the ruler is not the most universal icon. Little white dots or boxes that show up on the corner of the image when the image is selected would be a more common representation of the ability to resize. Additionally, the user should have more freedom to adjust the size rather than just the 3 options given when the ruler is selected. Additionally, a crop option for images is expected.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6	#4 Consistency and Standards #3 User Control and Freedom	2
Item 7	**Only one response can be submitted** to each discussion board prompt. There is no option to submit another response. The user receives the message "Your response has been submitted. Engage and discuss with other learners below!" Once posted, the user can delete their own post. However, this does not result in an an option to submit another new post as would be expected.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 --> after response is submitted	#4 Consistency and Standards #3 User Control and Freedom	2
Item 8	There is a grey "**view my response**" button after the user posts a discussion board response below the message: "Your response has been submitted. Engage and discuss with other learners below!" When the button is hovered over, it turns blue. When it is clicked, there is no visible change of system status. The button does not show that it has recognized the click, and when clicked, the button does not take the user to their own post as expected, but rather the top of the first page of responses. The only function it has is to highlight the user's response in light blue, but the user still has to scroll through pages of discussion board responses to find it. It should show that the interface has recognized the click, highlight the user's response blue, *and* take the user directly to their own post.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 --> after response is submitted	#1 Visibility of System Status #4 Consistency and Standards	2

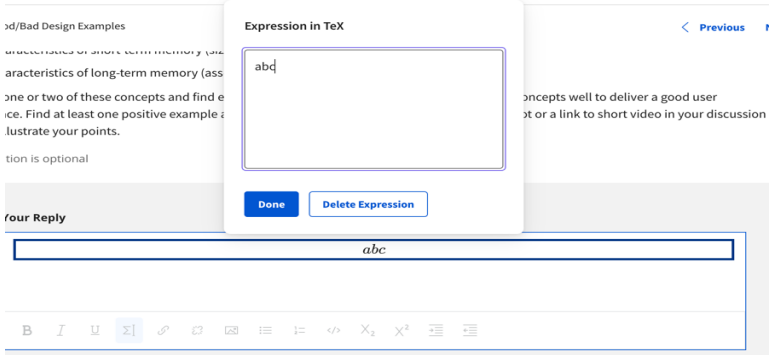
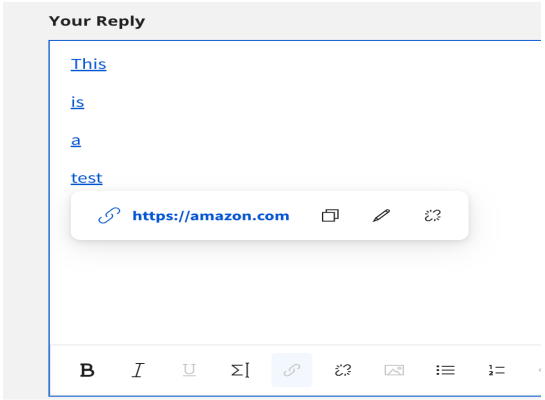
Item 9	When certain **equations** are entered, the equation textbox (example: photo 9b) is surrounding the text only versus in other equations like the italicized example 2a + 3b = c (9a)where the equation text box spans the length of the response window. This causes some confusion when a user tries to add another equation after. You can see in the example "this is regular text"(9c), additional equations can be added to the end of a line that begins (or ends) with regular text ("2ndEquation") (9c). Additional equations can also be added on a line starting with a bullet point. (9d) Another equation was also able to be added after the full length text box equation (9a). The new equation text box can only be placed down below since the equation text box surrounding 2a+3b=c is the full length of the window. The problem arises in the 3rd example (9e). To make this equation, I typed regular text and then highlighted it and clicked the math symbol. The equation text box is placed only around the text, not spanning the length of the window. This allows me to put my curser at the end of the text and click to add another equation(9f). However, when the equation button is clicked, the equation already on that line (abc) gets covered up/erased by the new equation box that spans the length of the window (9g). It also places the new equation centered, rather than on the left side (9h). There is no feature to change the pacement of text in the window to make it left, center, or right. This lack of consistency in equation textbox size and placement leaves the user likely confused and frustrated.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6	#1 Visibility of System Status #4 Consistency and Standards	2

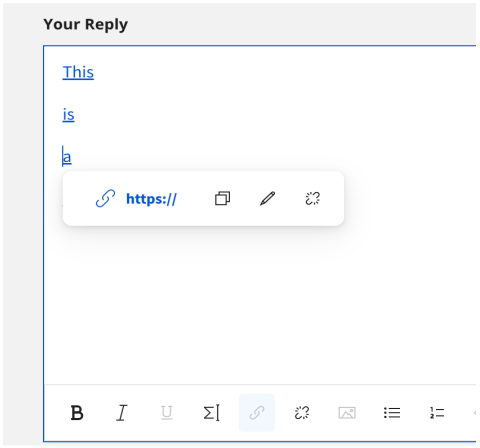
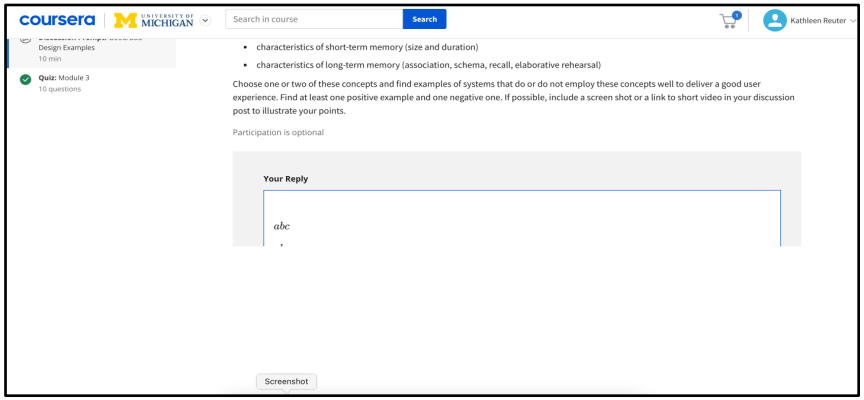
Item 10	When adding a **hyperlink**, if you highlight more than one paragraph of text, and create a valid hyperlink, the link attaches to the last paragraph of text only (10a). It still underlines and blues the other paragraphs that were highlighted so they look hyperlinked (10b), but when clicked on, they lead to an invalid/missing link. If you click on the missing link where it says "https://" there is no error message from the site letting you know that the link is missing/invalid. If the link can attach only to the last paragraph, the system should only chnage the color & underline the part of the text that was successfully hyperlinked as the common standard is that the blue-underlined visual signals that the text is "clickable".	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 --> text box hyperlink option	#1 Visibility of System Status #4 Consistency and Standards	2
Item 11	When analyzing the math equation feature, my discussion board page got stuck only partially scrolling. The bottom **half of the screen became white** (11a), blocking half of what is supposed to be seen on the screen. When I refrshed the page, it reset back to a normal full screen.	Discussion Prompt: Self-Reflection in Week 3	#4 Consistency and Standards	2
Item 12	In some discussion text boxes, the user is only able to go back and add one letter into text that has already been typed. After one letter is typed, the blinker is automatically moved back to the end of the text. This is not efficient nor consistent with expectations.	Discussion Prompt: Self Reflection Week 1	#4 Consistency and Standards #7 Flexibility and Efficiency of Use	2
Item 13	There is no **separation between discussion board posts** other than the upvote and reply button and timestamp. Perhaps a very thin horizontal line would allow for more separation and easy scanning of the posts while skimming over all posts. There should also be more of a differentiation in the "look" of an original post versus a reply to an original post.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6	#8 Aesthetic and Minimalist Design	1
Item 14	When scrolling through other users' responses, the **page length** varies and can sometimes get very long depending on the size/number of embedded images present. This makes it difficult to remember where certain posts are and to navigate reading through all the posts. There is also a "page" navigator at the bottom which is a very useful tool for users to have a checkpoint or reference for how many responses they are sifting through as well as an easy way to locate and return to certain posts at a later time. One might consider decreasing the number of responses on each page or limit each page to a certain length rather than a certain number of posts.	Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 --> after response is submitted	#8 Aesthetic and Minimalist Design	1

Photos				
Item 9a	9a 			
Item 9b				

Item 9c				
Item 9d				

Item 9e				
Item 9f				
Item 9g				

Item9h				
Item 10a				

Item 10b				
Item 11a				

Findings and Recommendations

Key Findings

Finding 1: Absence of undo/redo options

Severity: 3 out of 4
Location: Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6
Heuristic(s) Violated: *Consistency and Standards*

There are no **undo/redo** buttons. Most users expect there to be an undo/redo option when typing paragraphs of text with text formatting options such as bold, italicize, underline, bullet, etc. This should be an option as it would be consistent with standards provided on most discussion board interfaces as well as other typing-focused interfaces users typically have previous experience with such as Microsoft Office, Google Docs, etc.

Recommendation:
Include an undo button with a counter-clockwise arrow icon and a redo button with a clockwise arrow located on the bottom tool bar right before the B (Bold) button.



Finding 2: Saving text typed in text box

Severity: 3 out of 4
Location: Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6
Heuristic(s) Violated: *Error Prevention*

If you type text in the discussion prompt box and refresh the page or leave the discussion prompt page to go to a different part of the course, any text typed in the text box will not be saved from the previous visit. There should be a popup to prevent user errors asking if the user is sure they want to leave and warning them that their work will not be saved. Being able to navigate the website is important for users to be able to complete their intended tasks. In this case, users should be able to review information on other weeks/pages while constructing their discussion post to complete their assignments.

Recommendation:
Nielsen's 5th heuristic, *Error Prevention*, advises to either present users with a warning before they commit to losing their work completely, or change the design so that the issue no longer exists. Based on the nature of Coursera's purpose as a learning platform, I think it is important to provide the user with

seamless navigation as they return to previously learned content when completing assignments. The discussion board text box should keep the user's text saved automatically when they are logged into their account. That way, there is no risk of losing their work when navigating away from the page. Additionally, the user can begin to type a response and save it as a draft to be returned to at a later date.

Finding 3: Auto-condense replies on discussion board posts

Severity: 2 out of 4
Location: Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 → Discussion Board responses and replies
Heuristic(s) Violated: *Flexibility and Efficiency of Use*

When replying to a discussion board post, if the post has multiple replies already, the text box for submitting a reply appears at the end of the long thread of replies. A typical user might only want to read through the original post and share their thoughts on that. However, they must scroll past every reply to find the box where they can submit their own reply. Additionally, some users may be interested in only reading original posts, rather than reading through all the replies as well. Nielsen suggests in his 7th heuristic, *Flexibility and Efficiency of Use*, that the system should offer options for how a user would like to interact with the content. Additionally, the options should offer efficiency to both repeat and new users.

Recommendation:
The collection of replies following any given discussion board response should be automatically hidden with an option that says, "show x replies." There should be a text box available immediately following the original poster's text so the user can easily reference the original poster's thoughts as they form their own reply. This has the additional benefit of making scrolling/reading through responses more efficient. If a given response captures a user's attention, they can then choose to expand the replies to read the entire thread. You might also choose to allow the user to customize their reading experience. To do this, include a button at the top of each page of posts that says, "show all replies." When clicked, this button should change to perform the opposite action, "hide all replies."

Finding 4: Use consistent terminology when upvoting/liking a post

Severity: 2 out of 4
Location: Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 → Interacting with a discussion board post
Heuristic(s) Violated: *Consistency and Standards*

When interacting with discussion board responses, the term "x Upvotes" is not consistent with terminology used on most platforms. Nielsen's 4th heuristic highlights the importance of reducing the cognitive load by maintaining consistent language across user's platforms. Most users are familiar with the term "like." It is commonly coupled with an icon such as a heart or a thumbs-up. The word "like" is an actionable step the user could take. The term "x Upvotes" is a more passive phrase that connotates a statistic rather than an actionable step.

Recommendation:
Change the “x Upvotes” button with an up arrow icon to a "Like" button with a thumbs-up icon. Not only is the terminology more familiar to the user, but this would produce more consistency throughout the Coursera page as it appears at the bottom of all other items on the course website besides the Discussion Prompt page.

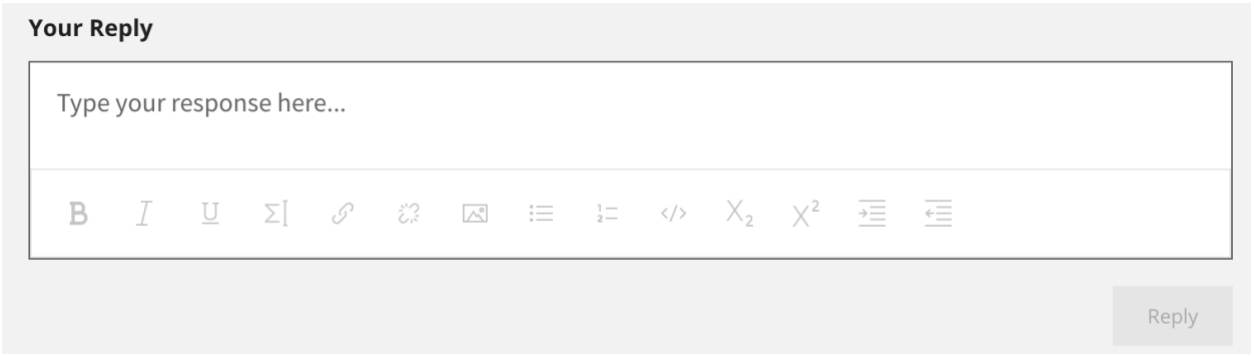


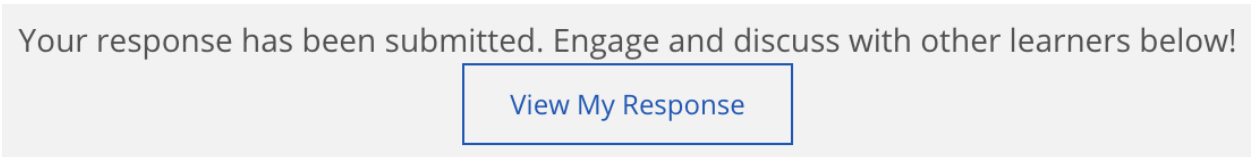
Finding 5: Use consistent terminology when responding to a discussion board prompt

Severity: 2 out of 4
Location: Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 → Below prompt description
Heuristic(s) Violated: *Consistency and Standards*

The word "reply" is confusing to a user when it comes to submitting an original post to the discussion board since they are not replying to anyone, but rather submitting an original thought in response to a prompt. Additionally, after clicking “Reply,” the user receives a message that states, "Your ***response*** has been submitted.” Use of the phrase “(Your) Reply” violates Nielsen’s 4th heuristic, *Consistency and Standards*.

Recommendation:
To maintain consistent terminology across the interface, "Submit Response" should be used when a user is responding directly to a prompt. “Submit Response” would be more familiar and consistent with interactions users have had with other discussion-board interfaces. It is also more consistent with the terminology used in the text box that says, “Type your response here…” and the feedback the user receives from the system once a response has been submitted. The word "reply" should be saved for when a user is submitting a reply to another user's initial response.



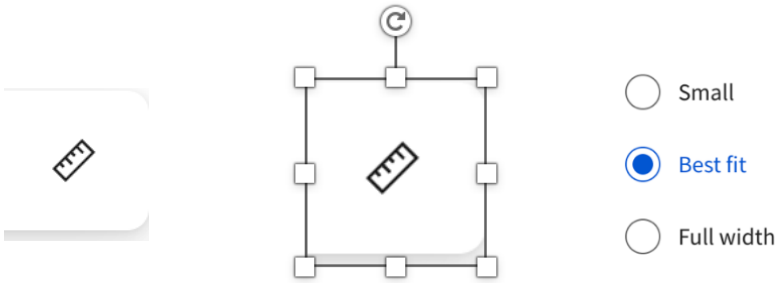


Finding 6: More user control when resizing an image

Severity: 2 out of 4
Location: Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 → Image tool bar menu when an image is inserted and selected
Heuristic(s) Violated: *Consistency and Standards*

When resizing an inserted image, the user must click on the image and then click on a ruler icon to see the photo sizing menu. There are only 3 size options and no ability to crop the photo internally.

Recommendation:
The ruler icon is not the most universal icon to signal to the user the ability to resize an image. Little white dots or boxes present on the corners of the image when the image is selected would be a more commonly recognized representation of the ability to resize. Additionally, the user should have more freedom to adjust the size rather than just the 3 options given when the ruler is selected. With the white dots, the user should be able to drag an image to a custom size of their liking. This offers the user more control and freedom while maintaining consistency with the user's expectations. Additionally, a crop feature for images is expected based on users' previous experiences sizing photos within most text-based interfaces.



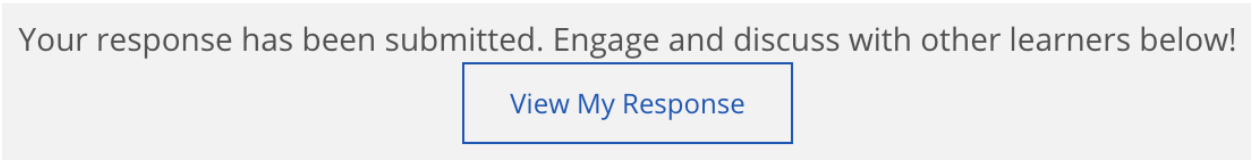
Finding 7: User can only submit one post, even after deleting their post

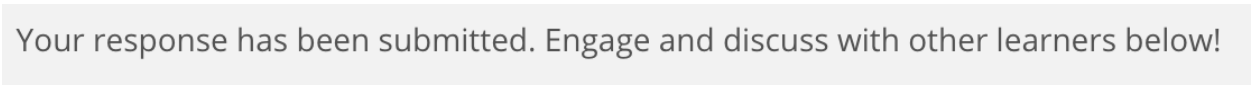
Severity: 2 out of 4
Location: Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 --> after response is submitted
Heuristic(s) Violated: *Consistency and Standards & Error Prevention*

Only one response can be submitted to each discussion board prompt. There is no option to submit another response. There is also no message to the user proclaiming this limitation. The user receives the message "Your response has been submitted. Engage and discuss with other learners below!" Once posted, the user can delete their own post. However, this does not result in an option to submit another new post as would be expected. There is no message to the user that they are not able to submit another post. Instead, the "View My Response" button simply disappears.

Recommendation:

Following Nielsen's 5th Heuristic of Error Prevention, the system should notify the user how many discussion board posts are allowed. Additionally, when the user clicks to delete their post, a message should be present stating that "a new attempt at posting on this discussion board will not become available." alongside the pop-up making sure the user intends to follow through with the deletion of their post as it cannot be undone. Another option would be to allow for multiple posts per user.





Finding 8: Trouble finding user's own post

Severity: 2 out of 4
Location: Discussion Prompt: Self-Reflection in Week 1, 2, 3, 4, & 5, and Heuristic Evaluation Sandbox in week 6 --> after response is submitted
Heuristic(s) Violated: *Visibility of System Status*

There is a grey "view my response" button after the user posts a discussion board response below the message: "Your response has been submitted. Engage and discuss with other learners below!" When the button is hovered over, it turns blue. When it is clicked, there is no visible change of system status. The button does not show that it has recognized the click. This violates Nielsen's heuristic Additionally, when the button is clicked, the button does not take the user to their own response as expected, but rather the top of the first page of responses. The only function it has is to highlight the user's response in light blue/grey, but this color is not a high enough contrast for the user to notice when scrolling and scanning quickly. Additionally, the user still has to scroll through pages of discussion board responses to find it.

Recommendation:
The interface should show that it has recognized the click by briefly turning the "view my response" button a different color (suggest white) when it is pressed down. It should still highlight the user's response in a brighter blue and then go the next step to take the user directly to their own post.

